

TED UNIVERSITY

CMPE 491

Senior Design Project I

Intelligent MRI Image Analysis and Automated Report Generation System

Project Proposal

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Project Description

Magnetic Resonance Imaging (MRI) is widely used to evaluate spinal disorders, including intervertebral disc herniation, disc degeneration, and spinal stenosis. However, interpreting MRI examinations requires radiologists to carefully review multiple image sequences and anatomical structures, making the reporting process time-consuming particularly in healthcare facilities with high imaging workloads. This project aims to develop an intelligent clinical decision-support system that automatically analyzes spine MRI images, identifies selected spinal abnormalities and generates structured draft radiology reports. The primary objective is to assist radiologists in interpreting MRI examinations more efficiently, reduce repetitive manual tasks and improve the consistency of radiological reporting.

The proposed system will utilize artificial intelligence, deep learning, and medical image processing techniques to analyze spine MRI scans and detect abnormalities such as disc herniation, disc degeneration, and spinal canal narrowing. Deep learning models, including convolutional neural networks and segmentation architectures such as U-Net, will be investigated and trained using suitable publicly available, annotated spine MRI datasets. The system will preprocess uploaded MRI images, identify relevant anatomical structures and abnormalities, visualize detected findings, and extract measurable information where supported by the models. Based on these findings, a report generation module will produce a structured draft radiology report using predefined medical templates and, where appropriate, a language model constrained by the detected findings. Through an interactive interface, radiologists will be able to review MRI images, examine highlighted abnormalities, edit the generated report and approve it before export. The system is intended to support, rather than replace, radiologists and will be developed as a research prototype to evaluate the feasibility of AI-assisted spine MRI interpretation and automated reporting.

Project Information

Project Name	Intelligent MRI Image Analysis and Automated Report Generation System
Project Web Page	https://mri-project-site.vercel.app
Supervisor	Dr. Mert Burkay ÇÖTELİ
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